```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import r2_score

import pickle
```

```
df = pd.read_csv("/content/car data.csv")
df.head()
```

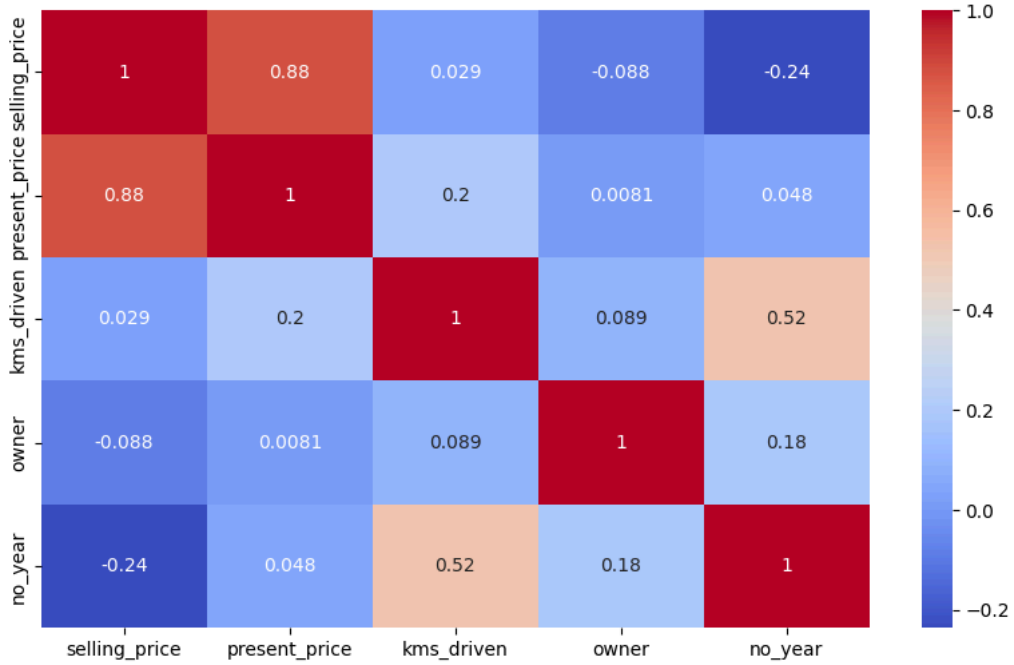
	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Transmission	Owner
0	ritz	2014	3.35	5.59	27000	Petrol	Dealer	Manual	0
1	sx4	2013	4.75	9.54	43000	Diesel	Dealer	Manual	0
2	ciaz	2017	7.25	9.85	6900	Petrol	Dealer	Manual	0
3	wagon r	2011	2.85	4.15	5200	Petrol	Dealer	Manual	0
4	swift	2014	4.60	6.87	42450	Diesel	Dealer	Manual	0

```
df.columns = df.columns.str.strip().str.lower()

current_year = 2024
df['no_year'] = current_year - df['year']

df.drop(['car_name','year'], axis=1, inplace=True)

df.isnull().sum()
```

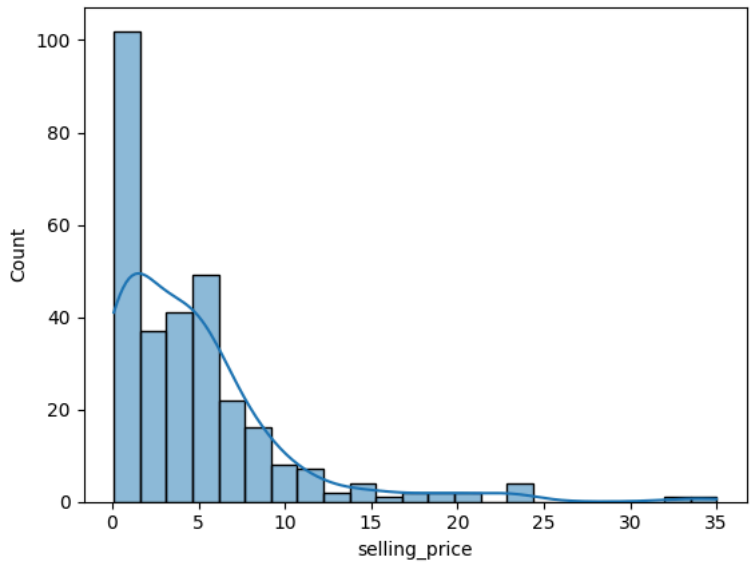
	0
selling_price	0
present_price	0
kms_driven	0
fuel_type	0
seller_type	0
transmission	0
owner	0
no_year	0

dtype: int64

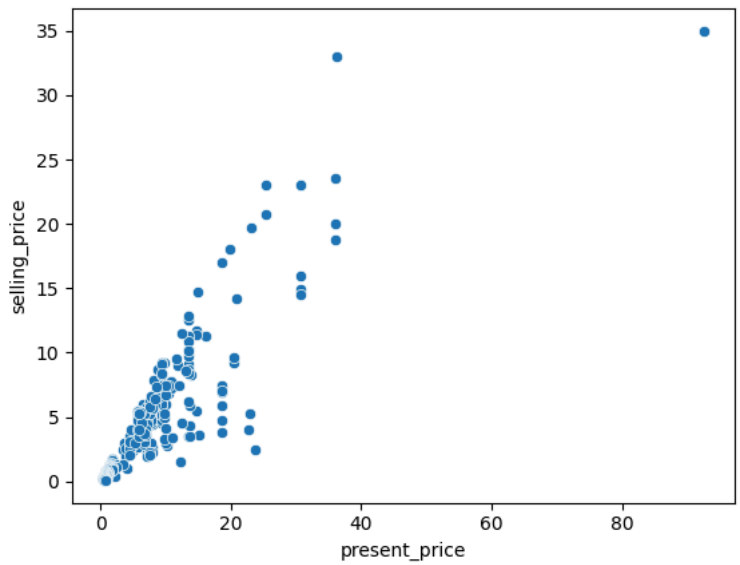
```
numeric_df = df.select_dtypes(include=['number'])

plt.figure(figsize=(10,6))
sns.heatmap(numeric_df.corr(), annot=True, cmap='coolwarm')
plt.show()
```




```
sns.histplot(df['selling_price'], kde=True)
plt.show()
```




```
sns.scatterplot(x=df['present_price'], y=df['selling_price'])
plt.show()
```



```
X = df.drop('selling_price', axis=1)
y = df['selling_price']
```

```
X = pd.get_dummies(X, drop_first=True)
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)
```

```
lr = LinearRegression()
lr.fit(X_train, y_train)

y_pred = lr.predict(X_test)
print("Linear Regression R2:", r2_score(y_test, y_pred))
```

```
Linear Regression R2: 0.848981302489908
```

```
from sklearn.ensemble import RandomForestRegressor

rf = RandomForestRegressor()
rf.fit(X_train, y_train)
```

```
▾ RandomForestRegressor
RandomForestRegressor()
```

```
pickle.dump(rf, open("car_model.pkl", "wb"))
```

```
rf = RandomForestRegressor()
rf.fit(X_train, y_train)

pickle.dump(rf, open("car_model.pkl", "wb"))
```

```
!ls
```

```
'car data.csv'   car_model.pkl   sample_data
```

Start coding or generate with AI.